

Exercise 08 :PhP

Create a simple Book Order website connected to a PHP backend.

Task 1 (10 Marks):

Create an HTML page named index.html that contains a form with the following fields:

Customer Name

Email

Book Title

Quantity

Submit Button

The form must send data to order.php using the POST method.

Task 2 (20 Marks):

Create order.php that:

Receives the submitted data.

Stores the values into PHP variables.

Displays all submitted information.

Task 3 (20 Marks):

Validate the form:

If any field is empty, display:

Please fill all fields.

Otherwise continue.

Task 4 (20 Marks):

Calculate the total price.

Book price:

£15

Formula:

Total = Quantity × 15

Display the total price.

Task 5 (15 Marks):

If Quantity is greater than or equal to 3, display:

You received a 10% discount.

Also display the final price after discount.

Task 6 (15 Marks):

Use CSS to make the form look clean and centred.

```
<!DOCTYPE html>
<html>
<head>
<!DOCTYPE html>
<html>
<head>
  <title>Book Order</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h1>Book Order Form</h1>
<form action="order.php" method="POST">
<input type="text" name="name" placeholder="Customer Name">
<input type="email" name="email" placeholder="Email">
<input type="text" name="book" placeholder="Book Title">
<input type="number" name="quantity" placeholder="Quantity">
<button type="submit"> Order </button>
</form>
</div>
</body>
</html>
```

order.php:

```
<?php $name = $_POST["name"]; $email = $_POST["email"]; $book = $_POST["book"];
$quantity = $_POST["quantity"];
if( empty($name) || empty($email) || empty($book) || empty($quantity) ){
  echo "<h2>Please fill all fields.</h2>";
}
else{
  $price = 15;
  $total = $quantity * $price;
  echo "<h2>Order Details</h2>";
  echo "Customer Name: $name <br>";
  echo "Email: $email <br>";
  echo "Book Title: $book <br>";
  echo "Quantity: $quantity <br>";
  echo "Total Price: £$total <br>";
  if($quantity >= 3){
    $discount = $total * 0.10;
    $finalPrice = $total - $discount;
    echo "<br>";
    echo "You received a 10% discount.<br>";
    echo "Final Price: £$finalPrice";
  }
}
?>
```

style.css:

```
body{
  font-family: Arial;
  background:#f2f2f2;
}
.container{
  width:400px;
  margin:50px auto;
  background:white;
  padding:20px;
  border-radius:10px;
}
input{
  width:100%;
  padding:10px;
  margin:10px 0;
}
button{
  width:100%;
  padding:10px;
  background:green;
  color:white;
  border:none;
}
```

Exercise 09 : Node.js

Create a Student Management System using React.js, Node.js, Express.js, and MySQL.

Part 1 – Database (20 Marks):

Create a MySQL database called student_db.

Create a table called students with the following columns:

id (INT, AUTO_INCREMENT, PRIMARY KEY)

name (VARCHAR(100))

email (VARCHAR(100))

major (VARCHAR(100))

Insert at least 3 sample students into the table.

Part 2 – Backend (40 Marks):

Using Node.js and Express.js, connect your application to MySQL.

Create the following REST API endpoints:

GET /students → Return all students.

GET /students/:id → Return one student by ID.

POST /students → Add a new student.

PUT /students/:id → Update a student.

DELETE /students/:id → Delete a student.

All requests and responses should use JSON.

Part 3 – Frontend (40 Marks):

Using React.js, create a page that:

Displays all students in a table.

Includes a form to add a new student.

Allows updating an existing student.

Allows deleting a student.

Uses Axios to communicate with the backend.

Database (MySQL)

```
CREATE DATABASE student_db;
USE student_db;
CREATE TABLE students (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100),
  major VARCHAR(100)
);
INSERT INTO students(name,email,major)
VALUES
('John Smith','john@gmail.com','Computer Science'),
('Sara Ali','sara@gmail.com','Information Technology'),
('Mike Brown','mike@gmail.com','Software Engineering');
```

server.js:

```
const express = require("express");
const mysql = require("mysql2");
const cors = require("cors");
const app = express();
app.use(cors());
app.use(express.json());
const db = mysql.createConnection({
  host: "localhost",
  user: "root",
  password: "",
  database: "student_db"
});
db.connect(err => {
  if(err) throw err;
  console.log("MySQL Connected");
});
app.get("/students", (req, res) => {
  db.query("SELECT * FROM students", (err, result) => {
    if(err) throw err;
    res.json(result);
  });
});
app.get("/students/:id", (req, res) => {
  db.query(
    "SELECT * FROM students WHERE id=?",
    [req.params.id],
    (err, result) => {
      if(err) throw err;
      res.json(result);
    }
  );
});
app.post("/students", (req, res) => {
  const {name, email, major} = req.body;
  db.query(
    "INSERT INTO students(name,email,major) VALUES(?,?,?)",
    [name, email, major],
    (err) => {
      if(err) throw err;
      res.json({message: "Student Added"});
    }
  );
});
app.put("/students/:id", (req, res) => {
```

```

const {name,email,major}=req.body;
db.query(
  "UPDATE          students          SET
name=?,email=?,major=? WHERE id=?",
  [name,email,major,req.params.id],
  (err)=>{
    if(err) throw err;
    res.json({message:"Student Updated"});
  }
);
});
app.delete("/students/:id",(req,res)=>{
  db.query(
    "DELETE FROM students WHERE id=?",
    [req.params.id],
    (err)=>{
      if(err) throw err;
      res.json({message:"Student Deleted"});
    }
  );
});
app.listen(5000,()=>{
  console.log("Server running on port 5000");
});

```

App.jsx:

```

import {useState,useEffect} from "react";
import axios from "axios";
function App(){
  const [students,setStudents]=useState([]);
  const [name,setName]=useState("");
  const [email,setEmail]=useState("");
  const [major,setMajor]=useState("");
  useEffect(()=>{
    loadStudents();
  },[]);
  const loadStudents={()=>{
    axios.get("http://localhost:5000/students")
    .then(res=>setStudents(res.data));
  }
  const addStudent={()=>{
    axios.post("http://localhost:5000/students",{
      name,email,major
    }).then(()=>{
      loadStudents();
    });
  }
  const deleteStudent=(id)=>{
    axios.delete(`http://localhost:5000/students/${id}`)
    .then(()=>{
      loadStudents();
    });
  }
  return(
    <div>
    <h1>Student Management System</h1>
    <input placeholder="Name" onChange={(e)=>setName(e.target.value)} />
    <input placeholder="Email" onChange={(e)=>setEmail(e.target.value)} />
    <input placeholder="Major" onChange={(e)=>setMajor(e.target.value)} />
    <button onClick={addStudent}> Add Student </button>
    <table border="1">
    <thead>
    <tr>

```

```

<th>ID</th>
<th>Name</th>
<th>Email</th>
<th>Major</th>
<th>Delete</th>
</tr>
</thead>
<tbody>
{
  students.map(student=>(
    <tr key={student.id}>
    <td>{student.id}</td> <td>{student.name}</td> <td>{student.email}</td>
    <td>{student.major}</td>
    <td>
      <button onClick={()=>deleteStudent(student.id)}> Delete </button>
    </td>
    </tr>
  ))
}
</tbody>
</table>
</div>
); }
export default App;

```